Theory of Elementary Particles: Lecture 5

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Symmetries in Particle Physics

Groups
- Basic definition
- Examples
- Representation

Lie Groups
- Example
- Unitary Lie Groups
- Commutator
- Lie Algebra
- Unitary group

Rank of the group and Casimir operators

SU(2)

Elements of the representation theory and Lie groups

- H. Georgi, *Lie algebras in particle physics*, Westview 1999.
- H. Lipkin, *Lie groups for pedestrians*, North-Holland Publishing Company, Amsterdam 1966.
- P. Cvitanovic, *Group theory*, Princeton University 2008.
- M. Hamermesch, *Group theory and its application to physical problems*, Dover publications 1989.

Symmetries of the Field Theory:

- Poincare group
- Global symmetry of QED: $U(1)$
- Gauge (local) symmetry of QED: $U(1)$
- Electro-weak interactions: gauge symmetry: $SU(2) \times U(1) \to U(1)$
- Strong interactions: $SU(3)$ gauge symmetry

Symmetry in particle classification: Quark Model

- Hadron wave functions → states of $SU(n)$ representations, where $n = 2, 3, 6$.

$$\Psi_{hadron} = \Psi_{colour} \times \Psi_{spin} \times \Psi_{flavour} \times \Psi_{space}.$$

- Baryon is a Fermion so its wave function must be antisymmetric
- Meson is a Boson so its wave function must be symmetric
- Space wave function → interaction between quarks: symmetry group $O(3)$.
- $SU_c(3)$ color singlet (totally antisymmetric)
- Spin-flavor (three quarks (u,d,s)):

$$\underbrace{SU(2)}_{spin} \times \underbrace{SU(3)}_{flavour} \subset SU(6).$$

Definition

Group is a set of elements, G, with a multiplication law satisfying requirements:

i) closure

$$a, b \in G \quad \Rightarrow a \cdot b \in G$$

let $a \cdot b \equiv a\,b$.

ii) associativity

$$a\,(b\,c) = (a\,b)\,c$$

iii) existence of unit element e

$$e\,a = a\,e = a$$

iv) existence of inverse element a^{-1} for every element $a \in G$

$$a a^{-1} = a^{-1} a = e$$

Unitary group: $U(n)$, $n \in \mathbb{N}$

- set $n \times n$ matrices such that if $U \in U(n)$ then $UU^\dagger = 1$
- $n = 1$: $U = e^{i\alpha}$, $\alpha \in \mathbb{R}$

Special unitary group: $SU(n)$, $n \in \mathbb{N}$

- set $n \times n$ matrices such that if $U \in U(n)$ then $UU^\dagger = 1$ and $\det U = 1$
- $SU(2)$ spin, isospin, rotations in 3-dim. space; homomorphic with $SO(3)$ (double cover: $SO(3) \cong SU(2)/\mathbb{Z}_2$)

Orthogonal group: $O(n)$, $n \in \mathbb{N}$

- set $n \times n$ matrices such that if $U \in O(n)$ then $UU^T = 1$

Special orthogonal group: $SO(n)$, $n \in \mathbb{N}$

- set $n \times n$ matrices such that if $U \in SO(n)$ then $UU^T = 1$ and $\det U = 1$

Scalar product invariant to $U(n)$, $SU(n)$ transformations

Let $(\psi_a)^* = \psi^a$, $a = 1, ..., n$ then if one assumes that

$$\psi'_a = U_{ab}\psi_b \tag{1}$$

and

$$\psi'^*_a = U^*_{ab}\psi^*_b \tag{2}$$

then the scalar product reads

$$\psi'^*_a\psi'_a = \psi^*_c U^*_{ac}U_{ad}\psi_d = \psi^*_c U^\dagger_{ca}U_{ad}\psi_d = \psi^*_c\delta_{cd}\psi_d = \psi^*_c\psi_c \tag{3}$$

Indeed the scalar product is invariant.

Remark

Einstein summation convention: we always assume summation over repeated indexes. The sum symbol $\sum$ will be usually omitted.

Direct product of groups G and H: $G \times H$

Given any two groups G and H, their direct product is a set $G \times H$ with multiplication law:

$$(g_1, h_1) \cdot (g_2, h_2) \equiv (g_1 g_2, h_1 h_2) \tag{4}$$

where $g_i \in G$, $h_i \in H$, $i = 1, 2$.

- $SU(2) \times U(1) \rightarrow$ electroweak interactions
- $SU(2) \times SU(3) \subset SU(6)$: spin-flavor symmetry group

Representation

Let G be a group and V a finite dimensional linear space. A map $a \to D(a)$ of the group G into the space $L(V)$ of linear operators (matrices) in V is called a representation of G in V if

$$D(ab) = D(a)D(b) \tag{5}$$

- Previously given definitions of groups $U(n)$, $SU(n)$,...etc. are given in terms of their defining representations.
- Dimension of a representation is just the dimension of the vector space on which it acts.

Invariant subspace

If there is a representation of G on V, a subspace of V, $U \subseteq V$ is called an invariant subspace if

$$u \in U \rightarrow D(a)u \in U \quad \forall a \in G \tag{6}$$

In other words

$$D(a)U \subseteq U \quad \forall a \in G \tag{7}$$

Notice that there exists always a trivial subspace spanned by $\vec{0}$ only

Irreducible representation

A representation is called irreducible if there are no nontrivial invariant subspaces.

In typical calculations

If matrix $D(a)$ can be put in block-diagonal form i.e. if there exists a non-singular matrix M independent of group elements, such that

$$MD(a)M^{-1} = \begin{pmatrix} D_1(a) & & 0 \\ & D_2(a) & \\ 0 & & \ddots \end{pmatrix} \forall a \in G$$

$D(a)$ is called reducible representation. It is given by a direct sum of irreducible representations

$$D(a) = D_1(a) \oplus D_2(a) \oplus ...$$

Notice that for every $D_i(a)$s exists corresponding invariant linear subspace.

Conditions for Lie Group

- Let $a(\boldsymbol{\theta}) = a(\theta_1, \theta_2, ..., \theta_n)$ be a group of elements labeled by n continuous real parameters.
- The identity element is is defined as

$$e \equiv a(0) \tag{8}$$

- The group multiplication

$$a(\boldsymbol{\theta})a(\boldsymbol{\phi}) = a(\boldsymbol{\xi}) \tag{9}$$

corresponds to the mapping of the parameter space on to itself

$$\mathbf{f}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \boldsymbol{\xi}, \tag{10}$$

which satisfies the properties

$$\mathbf{f}(0, \boldsymbol{\theta}) = \mathbf{f}(\boldsymbol{\theta}, 0) = \boldsymbol{\theta}, \quad \mathbf{f}(\boldsymbol{\theta}, \mathbf{f}(\boldsymbol{\phi}, \boldsymbol{\xi})) = \mathbf{f}(\mathbf{f}(\boldsymbol{\theta}, \boldsymbol{\phi}), \boldsymbol{\xi}), \tag{11}$$

and $\mathbf{f}(\boldsymbol{\theta}, \boldsymbol{\theta}') = 0$ if $a(\boldsymbol{\theta})^{-1}$ is parametrized as $a(\boldsymbol{\theta}')$.

If the function $\mathbf{f}$ is analytic (or continuously differentiable) with respect to its variables then above requirement define **Lie Group**.

Remark

A Lie group is a **differentiable manifold**, where the group operations are compatible with the smooth structure of the manifold. A differentiable manifold is a type of manifold that is locally similar enough to a linear space to allow one to do calculus. Smooth differential manifolds are associated with **vector fields**. Therefore, if G is any group acting smoothly on the manifold, then it acts on the vector fields.

- Example rotations in $\mathbb{R}^3$: $SO(3)$:
  - $$a(\phi,\theta,\zeta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\zeta & \sin\zeta & 0 \\ -\sin\zeta & \cos\zeta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
  - Notice that
  $$a(\mathbf{0}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Suppose $a(\boldsymbol{\theta})$ is in close neighborhood of the neutral element e then one may use analyticity property and expand given element in Taylor series

$$a(\boldsymbol{\theta}) = a(0) + \left.\frac{\partial a(\boldsymbol{\theta})}{\partial \theta_k}\right|_{\boldsymbol{\theta}=0} \theta_k + ..., \tag{12}$$

where $\boldsymbol{\theta}$ is real.
Let us introduce the so-called Lie group generator:

$$X_k \equiv -i \left.\frac{\partial a(\boldsymbol{\theta})}{\partial \theta_k}\right|_{\boldsymbol{\theta}=0} \tag{13}$$

Then

$$a(\boldsymbol{\theta}) = a(0) + i\boldsymbol{\theta} \cdot \mathbf{X} + ... \tag{14}$$

- Define the commutator of two group elements $a(\phi)$ and $a(\theta)$, lying near the identity, as

$$** = a(\phi)a(\theta)a(\phi)^{-1}a(\theta)^{-1} \tag{15}$$

- According to the Lie group property $**$ must belongs to group and should be represented as an element $a(\xi)$, where $\xi = \xi(\phi, \theta)$.
- For small θ and ϕ we may expand ξ in Taylor series:

$$\xi^k(\phi,\theta) = A^k + B^k_j\phi_j + B'^k_j\theta_j + C^k_{nm}\theta_n\phi_m + C'^k_{nm}\phi_n\phi_m + C''^k_{nm}\theta_n\theta_m + \ldots \tag{16}$$

where

$$A^k = \xi^k(0,0), \quad B^k_j = \left.\frac{\partial\xi^k}{\partial\phi_j}\right|_{(0,0)} \quad B'^k_j = \left.\frac{\partial\xi^k}{\partial\theta_j}\right|_{(0,0)} \tag{17}$$

$$C^k_{nm} = \left.\frac{\partial^2\xi^k}{\partial\theta_n\partial\phi_m}\right|_{(0,0)}, \quad C'^k_{nm} = \frac{1}{2}\left.\frac{\partial^2\xi^k}{\partial\phi_n\partial\phi_m}\right|_{(0,0)}, \quad C''^k_{nm} = \frac{1}{2}\left.\frac{\partial^2\xi^k}{\partial\theta_n\partial\theta_m}\right|_{(0,0)} \tag{18}$$

Remark

$$a(0)a(\boldsymbol{\theta})a^{-1}(0)a^{-1}(\boldsymbol{\theta}) = ea(\boldsymbol{\theta})ea^{-1}(\boldsymbol{\theta}) = a(\boldsymbol{\theta})a^{-1}(\boldsymbol{\theta}) = a(0) \tag{19}$$

Hence

$$\boldsymbol{\xi}(0,\boldsymbol{\theta}) = \boldsymbol{\xi}(\boldsymbol{\theta},0) = 0 \tag{20}$$

- From $\boldsymbol{\xi}(0,\boldsymbol{\theta}) = \boldsymbol{\xi}(\boldsymbol{\theta},0) = 0$ we get

$$0 = A^k + B'^k_j\theta_j + C''^k_{nm}\theta_n\theta_m + \ldots \tag{21}$$

$$0 = A^k + B^k_j\theta_j + C'^k_{nm}\theta_n\theta_m + \ldots \tag{22}$$

  for any vector $\boldsymbol{\theta}$

- hence

$$A^k = B^k_l = B'^k_l = C'^k_{nm} = C''^k_{nm} = 0 \tag{23}$$

  As the result

$$\xi^k = C^k_{nm}\theta_n\phi_m + \ldots, \tag{24}$$

$$
\begin{aligned}
** \quad &\approx \quad \left(e + i(\phi\cdot\mathbf{X}) - \frac{1}{2}(\phi\cdot\mathbf{X})^2\right)\left(e + i(\theta\cdot\mathbf{X}) - \frac{1}{2}(\theta\cdot\mathbf{X})^2\right) \\
&\quad\ \left(e - i(\phi\cdot\mathbf{X}) - \frac{1}{2}(\phi\cdot\mathbf{X})^2\right)\left(e - i(\theta\cdot\mathbf{X}) - \frac{1}{2}(\theta\cdot\mathbf{X})^2\right) \qquad (25) \\
&= \quad \left(e + i(\theta\cdot\mathbf{X}) - \frac{1}{2}(\theta\cdot\mathbf{X})^2 + i(\phi\cdot\mathbf{X}) - (\phi\cdot\mathbf{X})(\theta\cdot\mathbf{X}) - \frac{1}{2}(\phi\cdot\mathbf{X})^2\right) \\
&\quad\ \times\left(e - i(\theta\cdot\mathbf{X}) - \frac{1}{2}(\theta\cdot\mathbf{X})^2 - i(\phi\cdot\mathbf{X}) - (\phi\cdot\mathbf{X})(\theta\cdot\mathbf{X}) - \frac{1}{2}(\phi\cdot\mathbf{X})^2\right) \qquad (26) \\
&= \quad e - i(\theta\cdot\mathbf{X}) - \frac{1}{2}(\theta\cdot\mathbf{X})^2 - i(\phi\cdot\mathbf{X}) - (\phi\cdot\mathbf{X})(\theta\cdot\mathbf{X}) - \frac{1}{2}(\phi\cdot\mathbf{X})^2 + i(\theta\cdot\mathbf{X}) \\
&\quad\ + (\theta\cdot\mathbf{X})^2 + (\theta\cdot\mathbf{X})(\phi\cdot\mathbf{X}) - \frac{1}{2}(\theta\cdot\mathbf{X})^2 + i(\phi\cdot\mathbf{X}) + (\phi\cdot\mathbf{X})(\theta\cdot\mathbf{X}) + (\phi\cdot\mathbf{X})^2 \\
&\quad\ - (\phi\cdot\mathbf{X})(\theta\cdot\mathbf{X}) - \frac{1}{2}(\phi\cdot\mathbf{X})^2 \qquad (27) \\
&= \quad e + (\theta\cdot\mathbf{X})(\phi\cdot\mathbf{X}) - (\phi\cdot\mathbf{X})(\theta\cdot\mathbf{X}) \qquad (28) \\
&\to \quad e + \theta_k\phi_n(X_kX_n - X_nX_k) \qquad (29)
\end{aligned}
$$

$$** = e + \theta_k \phi_n (X_k X_n - X_n X_k) \tag{30}$$

on the other hand

$$** = a(\boldsymbol{\xi}) = e + i\xi_l X_l = e + iC^l_{kn}\theta_k \phi_n X_l \tag{31}$$

then

Commutation relation

$$\boldsymbol{[X_n, X_k] = iC^l_{nk} X_l} \tag{32}$$

Lie algebra

$$[X_k, X_l] = iC^n_{kl} X_n \tag{33}$$

The C^n_{kl}s called the structure constants of the group, are a set of real numbers with

$$C^l_{jk} = -C^l_{kj}$$

Notice that the Lie algebra of $SU(n)$ group is denoted by $su(n)$.

Jacobi identity

$$[X_j, [X_k, X_l]] + [X_k, [X_l, X_j]] + [X_l, [X_j, X_k]] = 0 \tag{34}$$

(exercise)

Comments

- Every Lie group has an associated **Lie Algebra** which is a **vector space** (tangent space) around the identity element of the group.
- A vector space is a structure "above" a field. There are 2 allowed operations - addition and scalar multiplication.
- Notice that the Lie algebra can be considered as a **linearization** of the Lie group (near the identity element).
- Around unit element:

$$\exp : \mathbf{g} \to G \tag{35}$$

  where $\mathbf{g}$ denotes the Lie algebra of the group
- Lie groups with unitary representations

$$a(\boldsymbol{\theta}) = \exp(i\boldsymbol{\theta} \cdot \mathbf{X}) \tag{36}$$

Adjoint representation

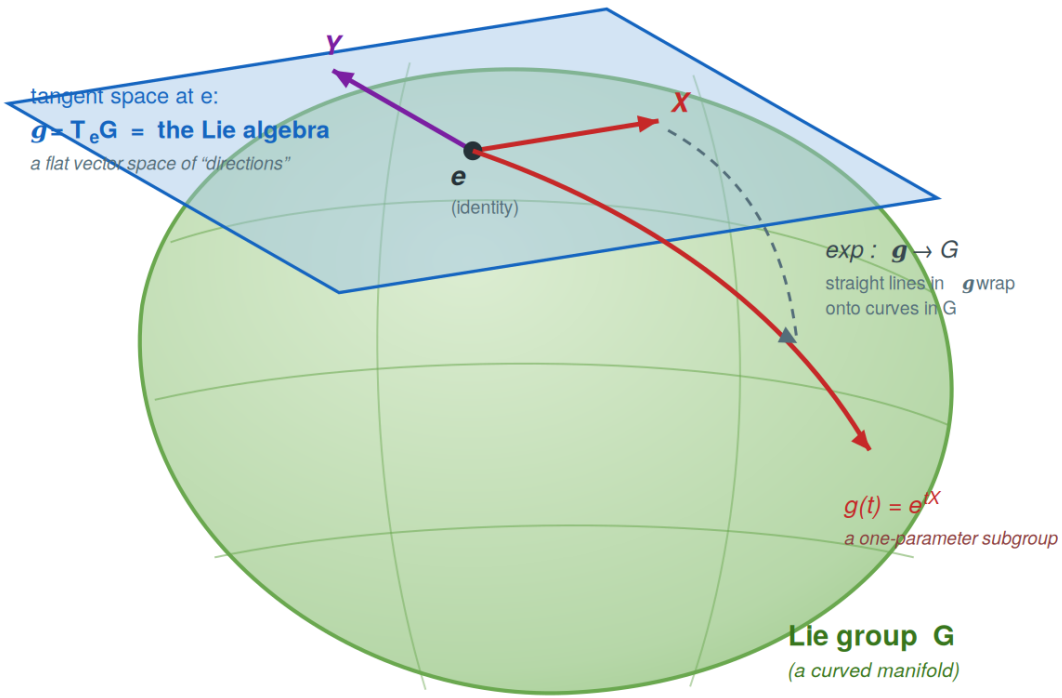
- We define a set of matrices

$$(T_j)^m_k \equiv -i\, C^m_{jk} \tag{37}$$

  which also satisfy (33) — by the Jacobi identity the structure constants themselves provide a representation of the algebra, the **adjoint** one.
- In the case of $SU(n)$ group the adjoint representation has dimension $n^2 - 1$

The Lie Algebra as a Tangent Space

the group is a curved manifold; its algebra is the flat space of directions at the identity

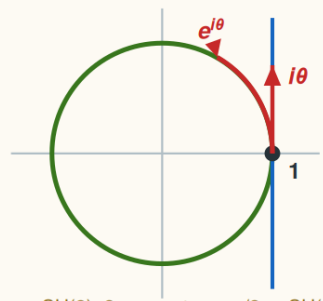


The dictionary

group G	algebra $\mathfrak{g}$
finite transformation	infinitesimal generator
$g = \exp(i\varepsilon_a T_a)$	$T_a = -i\,\partial g/\partial\varepsilon_a\,\vert_0$
curvature of G	$[T_a, T_b] = i f_{abc} T_c$
dim G	number of generators

the commutator lives in $\mathfrak{g}$: the algebra "remembers" how the group fails to commute

Example: U(1) = the circle



the tangent line at 1 is the imaginary axis:

$\mathfrak{g} = \{ i\theta \} = \mathfrak{u}(1)$

exp wraps it onto the circle: $e^{i\theta}$

SU(2): 3 generators $\tau_a/2$; SU(3): 8 generators $\lambda_a/2$

dim g = dim G ; the algebra determines the group near the identity — and that is all perturbation theory ever needs

Unitary groups

- Any unitary matrix $U \in U(n)$ can be represented as

$$U = \exp(iH), \tag{38}$$

  where H is hermitian ($\rightarrow$ exercise)
- If $\det U = 1$ we can choose H traceless, because $\mathrm{Tr} H = 2\pi k$, $k \in Z$.
- There are $n^2 - 1$ hermitian traceless basis matrices $n \times n$ H_j, linearly independent, so that

$$U = \exp\left[i \sum_{j=1}^{n^2-1} \alpha_j H_j \right] \tag{39}$$

Definition

Exponential function is defined by series

$$\exp(A) = 1 + A + \frac{A^2}{2!} + \ldots + \frac{A^n}{n!} + \ldots = \sum_{n=0}^{\infty} \frac{A^n}{n!} \tag{40}$$

$SU(2)$ group

$$U(\epsilon_1, \epsilon_2, \epsilon_3) = \exp\left(i\epsilon_a \frac{\tau_a}{2}\right), \tag{41}$$

$$\tau_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \tau_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \ \tau_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Example

Notice that $\tau_3^2 = 1$

$$\exp\left(i\epsilon_3 \frac{\tau_3}{2}\right) = 1 - \frac{(\epsilon_3/2)^2}{2!} + \frac{(\epsilon_3/2)^4}{4!} - \ldots + i\epsilon_3 \frac{\tau_3}{2} - \frac{i}{3!}\left(\epsilon_3 \frac{\tau_3}{2}\right)^3 + \ldots \quad (42)$$

$$= \cos\left(\frac{\epsilon_3}{2}\right) + i\,\tau_3 \sin\left(\frac{\epsilon_3}{2}\right) \quad (43)$$

$$= \begin{pmatrix} \cos\left(\frac{\epsilon_3}{2}\right) + i\sin\left(\frac{\epsilon_3}{2}\right) & 0 \\ 0 & \cos\left(\frac{\epsilon_3}{2}\right) - i\sin\left(\frac{\epsilon_3}{2}\right) \end{pmatrix} \quad (44)$$

$$= \begin{pmatrix} \exp\left(i\frac{\epsilon_3}{2}\right) & 0 \\ 0 & \exp\left(-i\frac{\epsilon_3}{2}\right) \end{pmatrix} \quad (45)$$

Complex conjugate representation

- If $D(a)$s form a representation of group, the elements $D^*(a)$s form complex conjugate representation since $D(a)D(b) = D(a\,b)$ implies $D^*(a)D^*(b) = D^*(a\,b)$ then in the case of the Lie algebra we have complex conjugation representation of the same structure with generators:

$$T_j \to -T_j^* \tag{46}$$

because $\exp(i\theta_a T_a) \to \exp(-i\theta_a T_a^*)$

Real representation

If T_j and $-T_j^*$ are equivalent i.e. there exists a nonsingular matrix S such that

$$ST_jS^{-1} = -T_j^* \quad \forall j \tag{47}$$

then T_j is called real representation.

$SU(2)$

Irreducible representations of $SU(2)$ are real (exercise).

* important property for the construction of the flavor wave function of mesons

** In reality it is real (for integer j), and pseudoreal, quaternionic (for rational j)

Rank of the $SU(n)$ group

The maximum number of mutually commuting generators in an $su(n)$ Lie algebra is called rank of the group. The rank of $SU(n)$ is $n-1$.
(exercise)

Statement

The Casimir operators and the $n-1$ mutually commuting generators are used to enumerate the states.

$SU(2)$ group

see e.g. J. Rzewuski, *Introduction to quantum theory*

- Lie algebra

$$[J_a, J_b] = i\epsilon_{abc} J_c, \tag{48}$$

  where $\epsilon_{123} = 1$.

- Example: Pauli matrices

$$J_i = \frac{\tau_i}{2} \tag{49}$$

  where the τ_as are 2×2 hermitian matrices.

- It is easy to show that $SU(2)$ is of rank 1.
- Define the Casimir operator which commutes with all the generators of given group

$$J^2 = J_1^2 + J_2^2 + J_3^2 \tag{50}$$

- We see that in order to label the states of the irreducible $SU(2)$ representation we may use the Casimir operator J^2 and one of the generators, e.g. J_3 so the representation contains the states $\mid \lambda, m\rangle$ so that

$$J^2 \mid \lambda, m\rangle = \lambda \mid \lambda, m\rangle, \tag{51}$$

$$J_3 \mid \lambda, m\rangle = m \mid \lambda, m\rangle \tag{52}$$

- define raising and lowering operators

$$J_{\pm} = J_1 \pm iJ_2 \tag{53}$$

Notice that

$$J^2 = \frac{1}{2}\left(J_+J_- + J_-J_+\right) + J_3^2 \tag{54}$$

as well as

$$\left[J_+, J_-\right] = 2J_3, \quad \left[J_{\pm}, J_3\right] = \mp J_{\pm} \tag{55}$$

- The states $J_{\pm} \mid \lambda, m\rangle$ are also the eigenstates of J_3 with eigenvalues $m \pm 1$

$$J_3J_+ \mid \lambda, m\rangle = \left(J_+J_3 + J_+\right) \mid \lambda, m\rangle = (m+1)J_+ \mid \lambda, m\rangle \tag{56}$$

so

$$J_+ \mid \lambda, m\rangle \sim \mid \lambda, m+1\rangle \tag{57}$$

- Notice that

$$J_\pm \mid \lambda, m\rangle = C_\pm(\lambda, m) \mid \lambda, m \pm 1\rangle \tag{58}$$

where $C_\pm(\lambda, m)$s are constants

-

$$\langle \lambda, m \mid J^2 - J_3^2 \mid \lambda, m\rangle = \langle \lambda, m \mid J_1^2 + J_2^2 \mid \lambda, m\rangle = \lambda - m^2 \geq 0 \tag{59}$$

hence $\lambda \geq m^2$

- Notice that

$$J_- J_+ = (J_1 - iJ_2)(J_1 + iJ_2) = J_1^2 + J_2^2 + iJ_1J_2 - iJ_2J_1 = J_1^2 + J_2^2 - J_3 \tag{60}$$

- Let j be the largest value of m then $J_+ \mid \lambda, m\rangle = 0$ and

$$0 = J_- J_+ \mid \lambda, j\rangle = (J^2 - J_3^2 - J_3) \mid \lambda, j\rangle = (\lambda - j^2 - j) \mid \lambda, j\rangle \tag{61}$$

$$\Rightarrow \quad \lambda = j(j+1) \tag{62}$$

- similarly if j' is the smallest value of m then $\lambda = j'(j'-1)$ because one needs to consider $J_+ J_- = J^2 - J_3^2 + J_3$
- $\Rightarrow j'(j'-1) = j(j+1) \Rightarrow (j'-j-1)(j'+j) = 0$ either $j' = j+1$ (in contradiction with the assumption that j is the largest value of m) **or $\mathbf{j' = -j}$**;

- Since J_- lowers m by one unit then $j - j' = 2j = n \in \mathbb{N}$ must be integer so

$$j = 0, \frac{1}{2}, 1, \frac{3}{2}, ... \tag{63}$$

and

$$\lambda = 0, \frac{3}{4}, 2, \frac{15}{4}, ... \tag{64}$$

- It is easy to show that

$$C_\pm(j,m) = \sqrt{(j \mp m)(j \pm m + 1)} \tag{65}$$

- For fixed value of j we have the finite-dimensional irreducible representation of $SU(2)$ with $2j + 1$ states $m = j, j - 1, ..., -j$

$$J_3 \mid j,m\rangle = m \mid j,m\rangle \tag{66}$$

$$J_\pm \mid j,m\rangle = C_\pm(j,m) \mid j,m \pm 1\rangle \tag{67}$$

- Example $j = \frac{1}{2}$, $m = \pm\frac{1}{2}$

$$J_3 \mid \frac{1}{2}, \pm\frac{1}{2}\rangle = \pm\frac{1}{2} \mid \frac{1}{2}, \pm\frac{1}{2}\rangle \tag{68}$$

if

$$\mid \frac{1}{2}, \frac{1}{2}\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \mid \frac{1}{2}, -\frac{1}{2}\rangle = \begin{pmatrix}0\\1\end{pmatrix} \rightarrow J_3 = \frac{1}{2}\begin{pmatrix}1 & 0\\0 & -1\end{pmatrix} J_+ = \begin{pmatrix}0 & 1\\0 & 0\end{pmatrix} J_- = \begin{pmatrix}0 & 0\\1 & 0\end{pmatrix} \tag{69}$$

and

$$J_1 = \frac{1}{2}\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} \quad J_2 = \frac{1}{2}\begin{pmatrix}0 & -i\\i & 0\end{pmatrix} \tag{70}$$

- Example (exercise): three-dimensional representation $j = 1$, so $m = 1, 0, -1$

$$J_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & 1 & 0\\1 & 0 & 1\\0 & 1 & 0\end{pmatrix}, \quad J_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}0 & -i & 0\\i & 0 & -i\\0 & i & 0\end{pmatrix}, \quad J_3 = \begin{pmatrix}1 & 0 & 0\\0 & 0 & 0\\0 & 0 & -1\end{pmatrix} \tag{71}$$